

Roll No.

TEE-501

**B. TECH. (EE) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

POWER SYSTEM—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

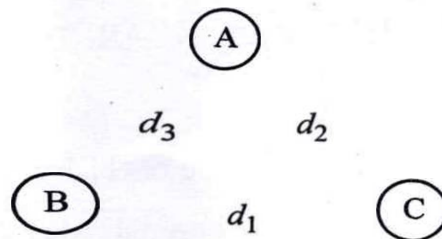
1. (a) Discuss Kelvin's law to find most economical size of the conductor diameter. (CO1, CO2)
- (b) A 2-conductor cable 1 km long is required to supply a constant current of 200 A throughout the year. The cost including installation is ₹ $(20a + 20)$ per meter where 'a' is the area of cross section in cm^2 . The cost of energy is 5P/kWh and interest and depreciation charge amount to 10 %. Calculate the most economical conductor size. Assume resistivity of conductor material to be $1.73 \mu\Omega \text{ cm}$. (CO1, CO2)

P. T. O.

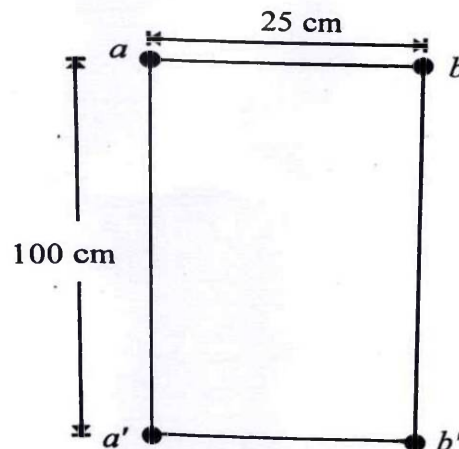
- (c) Compare the volume of the conductor material required in 1-phase two ac system with mid-point earth and dc two wire system with mid-point earth. (CO1, CO2)
2. (a) What do you mean by string efficiency ? Derive the expression of string efficiency for three-disc systems. (CO3)
- (b) Each line of a 3-phase system is suspended by a string of three identical insulators of self-capacitance C farad. The shunt capacitance of connecting metal work of each insulator is $0.2 C$ to earth and $0.1 C$ to line. Calculate the string efficiency of the system if a guard ring increases the capacitance to the line of metal work of the lowest insulator to $0.3 C$. (CO3)
- (c) The three bus-bar conductors in an outdoor substation are supported by units of post type insulators. Each unit consists of a stack of 3 pin type insulators fixed one on the top of the other. The voltage across the lowest insulator is 13.1 kV and that across the next unit is 11 kV . Find the bus-bar voltage of the station. (CO3)
3. (a) Discuss the theory of corona formation in high voltage transmission line. (CO6)
- (b) Derive the expression for the mechanical sag for the equal support. (CO6)
- (c) An overhead line has a span of 150 m between level supports. The conductor has a cross-sectional area of 2 cm^2 . The ultimate strength is 5000 kg/cm^2 and safety factor is 5. The specific gravity of the material

is 8-9 gm/cc. The wind pressure is 1.5 kg/m. Calculate the height of the conductor above the ground level at which it should be supported if a minimum clearance of 7 m is to be left between the ground and the conductor. (CO6)

4. (a) Derive the value of capacitance for three phase-three wire system as shown in figure below : (CO5)



- (b) Deduce the expression of inductance for three phase three wire lines which are symmetrically spaced at distance ' d '. Assume the radius of the conductor is ' r '. (CO5)
- (c) Two conductors of a single-phase line, each of 1 cm diameter, are arranged in a vertical plane with one conductor mounted 1 m above the other, as shown in the figure below. A second identical line is mounted at the same height as the first and spaced horizontally 0.25 m apart from it. The two upper and the two lower conductors are connected in parallel. Determine the inductance per km of the resulting double circuit line : (CO5)



5. (a) Discuss the construction and different layers of the underground cable.

(CO4, CO6)

- (b) A 33 kV, 50 Hz, 3-phase underground cable, 4 km long uses three single core cables. Each of the conductor has a diameter of 2.5 cm and the radial thickness of insulation is 0.5 cm.

Determine : (i) capacitance of the cable/phase (ii) charging current/phase (iii) total charging KVAR. The relative permittivity of insulation is 3.

(CO4, CO6)

- (c) A single core cable for use on 11 kV, 50 Hz system has conductor area of 0.645 cm^2 and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find : (i) the maximum electrostatic stress in the cable (ii) minimum electrostatic stress in the cable (iii) capacitance of the cable per km length (iv) charging current.

(CO4, CO6)

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B. TECH. (EE) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023

CONTROL SYSTEMS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

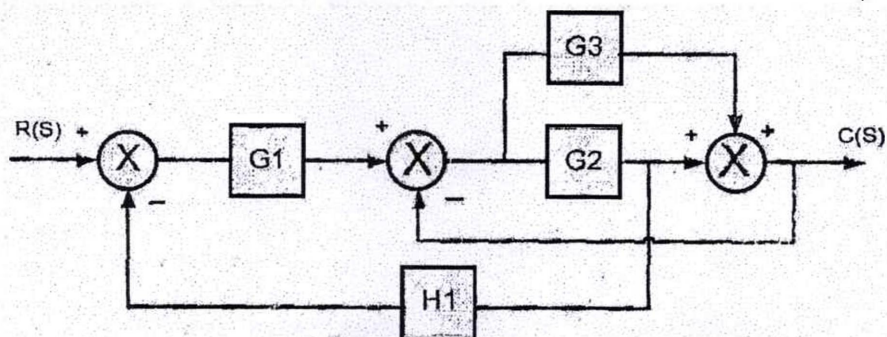
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the closed loop controls system and explain any one example with suitable block diagram.

(b) Using block reduction technique calculate transfer function of the block diagram given below : (CO1, CO2)



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- (c) Explain the effect of ξ on second order system performance for unit step input when : (CO1, CO2)

(i) $\xi = 0$

(ii) $0 < \xi < 1$

(iii) $\xi = 1$

(iv) $1 < \xi < \infty$

2. (a) Discuss the time domain specifications and mark them on the response of 2nd order system when subjected to unit step input.
- (b) Find K_p , K_v , and K_a and steady state error for a system with open loop transfer function as :

$$G(s)H(s) = \frac{10(s+2)(s+3)}{s(s+1)(s+5)(s+4)}$$

where the input is $r(t) = 3 + t + t^2$.

- (c) The open loop T. F. of certain unity feedback system is : (CO1, CO2)

$$G(s)H(s) = \frac{K(s+10)(s+20)}{s^2(s+6)}$$

Determine :

- (i) Range of K for stability
- (ii) Marginal value of K
- (iii) Location of roots for marginal stability

3. (a) Sketch the root locus plot and give the stability conditions for the system when open loop transfer function is given by : (CO3, CO4)

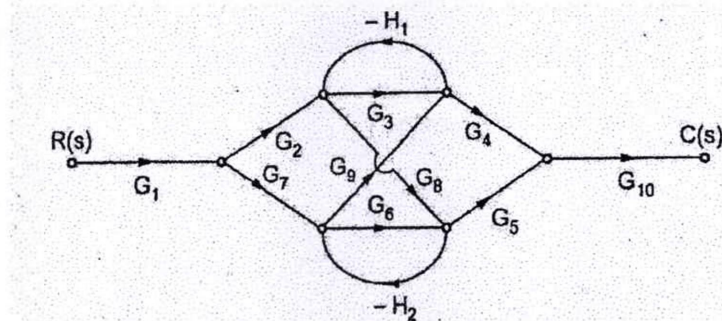
$$G(s)H(s) = \frac{K}{s(s+2)(s+4)}$$

- (b) Draw the Nyquist contour and explain the Nyquist stability criterion.
 (c) Calculate the transfer function from the given state model :

$$\mathbf{X}(t) = \begin{bmatrix} -3 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t) \text{ and } y(t) = [1 \quad 1] \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

4. (a) Calculate state transition matrix for $A = \begin{bmatrix} 0 & -1 \\ 2 & -3 \end{bmatrix}$ with the help of Laplace transform method. (CO5)

- (b) Using Mason's gain formula obtain the transfer function of the signal flow graph given below :



- (c) Judge the controllability and observability of a system with :

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$C = [10 \ 5 \ 1]$$

5. (a) Describe all the rules of Block reduction method. (CO4, CO6)
- (b) Discuss what you mean by settling time. A feedback control system has :

$$G(s)H(s) = \frac{15}{(s+1)(s+3)}$$

and $H(s) = 1$. Find peak-time, rise-time, maximum overshoot, and settling-time.

- (c) Determine the value of K such that the roots of the characteristic equation given below lie to the left of line $s = -1$.

$$s^3 + 10s^2 + 18s + K = 0$$

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TOE-502

B. TECH. (EE) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

COMPUTER BASED NUMERICAL AND STATISTICAL TECHNIQUES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the need of numerical techniques. (CO1)
- (b) The discharge Q over a notch for head H is calculated by the formula $Q = kH^{5/2}$, where k is a given constant. If the head is 75 cm and an error of 0.15 cm is possible in its measurement, estimate the percentage error in computing the discharge. (CO1)
- (c) Round-off the numbers 865250 and 37.46235 to four significant figures and compute e_a, e_r, e_p in each case. (CO1)

P. T. O.

2. (a) Find a real root of the equation $3x + \sin x - e^x = 0$ by the method of false position correct to four decimal places by taking 0.2 and 0.3 as initial values. (CO2)
- (b) Find a real root of the equation $x = e^{-x}$ using the Newton-Raphson method. (CO2)
- (c) Perform four iterations for finding out the roots of the equation $\cos x = 3x - 1$ using iteration method. (CO2)
3. (a) What is interpolation ? Discuss various operators associated with it. (CO3)
- (b) Using Gauss' backward interpolation formula, find the population for the year 1936 given that : (CO3)

Year	Population (in thousands)
1901	12
1911	15
1921	20
1931	27
1941	39
1951	52

- (c) Find the cubic Lagrange's interpolating polynomial from the following data : (CO3)

x	$f(x)$
0	2
1	3
2	12
5	147

4. (a) Discuss and derive Trapezoidal rule for numerical integration. (CO4)
 (b) Find dy/dx at $x = 6$, given that : (CO4)

X	Y
4.5	9.69
5.0	12.90
5.5	16.71
6.0	21.18
6.5	26.37
7.0	32.34
7.5	39.15

- (c) The speed, v metres per second, of a car, t seconds after it starts, is shown in the following table : (CO4)

t	v
0	0
12	3.60
24	10.08
36	18.90
48	21.60
60	18.54
72	10.26
84	5.40
96	4.50
108	5.40
120	9.00

Using Simpson's rule, find the distance travelled by the car in 2 minutes.

5. (a) Differentiate between the following : (CO5, CO6)
- (i) Initial value and boundary value problems
 - (ii) Single step and Multi-step methods for solution of differential equation

(5)

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- (b) Employ Picard's method to find $y(0.2)$, given that : (CO5, CO6)

$$\frac{dy}{dx} = 1 + y^2 \text{ and } y(0) = 0.$$

- (c) Solve the equation $\frac{dy}{dx} = x + y$ with initial condition $y(0) = 1$ by Runge-

Kutta's rule, from $x = 0$ to $x = 0.4$ with $h = 0.1$. (CO5, CO6)



Roll No.

TEE-503

**B. TECH. (EE) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

MICROPROCESSORS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) (i) How the multiplexed address and data buses (AD_0 - AD_7) get demultiplexed ? (CO1)
(ii) How are the control signals generated ?
- (b) Draw and explain the hardware model and programming model of 8085 microprocessor. (CO1)
- (c) Explain the following instructions of 8085 : (CO1)
 - (i) XCHG
 - (ii) JMP

P. T. O.

(iii) RET

(iv) DAA

(v) RAR

2. (a) Draw the timing diagram of MOV A, M. (CO2)

(b) The following program has a delay subroutine located at location 2060H. Read the program and answer the questions given at the end of the program : (CO2)

Memory locations	Mnemonics
2000H LXI SP, 20CDH	: Main Program
2003H LXI H, 0008H	
2006H MVI B, 0FH	
2008H CALL 2060H	
200BH OUT 01H	
DCR B	

CONTD

2060H PUSH H	: Delay Subroutine
2061H PUSH B	
MVI B, 05H	
LXI H, COUNT	
POP B POP H RET	

- (i) When the execution of the CALL instruction located at 2008H – 200AH is completed, list the contents stored at 20CCH and 20CBH the contents of the program counter, and the contents of the stack pointer.
- (ii) List the stack locations and their contents after the execution of the instructions PUSH H and PUSH B in the subroutine.

- (iii) List the contents of Stack pointer register after the execution of the instruction PUSH B located at 2061 H.
 - (iv) List the contents of Stack pointer register after the execution of instruction RET in the subroutine.
 - (c) Write an assembly language program of 8085 microprocessor for multiplication of two 8 bit numbers.
3. (a) Explain the following pins of 8086 : (CO3)
- (i) QS_0
 - (ii) MN/MX'
 - (iii) READY
 - (iv) $TEST'$
 - (v) RESET
- (b) What do you understand by minimum mode and maximum mode in 8086 ? Explain the functions of pins in minimum mode. (CO3)
- (c) Explain the addressing modes in 8086 microprocessor for control transfer instructions along with examples. (CO3)
4. (a) Explain the following instructions of 8086 microprocessor : (CO4)
- (i) AAA
 - (ii) SAR
 - (iii) LAHF
 - (iv) LOOP
 - (v) STD
- (b) (i) Draw and explain the block diagram of 8255, Programmable Peripheral Interface. (CO4)

- (ii) What will be the control word format for Mode 0 to configure port A and port Cu as output ports and port B and port CL, as input ports ?
- (iii) What will be the BSR control word format to set bit PC7 and reset bit PC3 ?
- (c) What is the function of priority resolver and interrupt service routine in 8259 programmable Interrupt controller ? (CO4)
5. (a) Write a program for 8086 microprocessor to find count of even numbers in an array of 8 bit numbers starting from 0501H offset in the segment 0200H. Size of array is stored at 0500 H offset address. Store result at offset 0400H in the same segment. (CO5, CO6)
- (b) What is DMA controller ? How does it reduce the burden of CPU ? Explain with the help of block diagram of 8257. (CO5, CO6)
- (c) Explain 8253/8254 programmable interval timer with block diagram. (CO5, CO6)

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TEE-504

**B. TECH. (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023
ELECTRICAL MACHINE DESIGN**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Derive the output equation of a single phase and three phase core type transformer. (CO1, CO3)
- (b) Differential between power and distribution transformer and explain the importance of window space factor. (CO1, CO3)
- (c) Enumerate the importance of No-load current of transformer with suitable diagram. (CO1, CO3)
2. (a) Explain in steps how to separate D and L for a DC machine. (CO2, CO4)

P. T. O.

- (b) Derive the output power of synchronous generator incorporating specific electric and magnetic loading. (CO2, CO4)
- (c) Determine the main dimensions of a 500 kVA, 50 Hz 3 phase alternator to run at 375 rpm. The average air gap flux density is 0.55 wb/mm^2 , the specific electric loading is 25000 A/m. The peripheral speed should not exceed 35 m/sec. (CO2, CO4)
3. (a) Explain the procedure for separation of D and L from DL product while designing induction motors. (CO4, CO6)
- (b) How choice of power factor and efficiency are considered in designing induction motor ? (CO4, CO6)
- (c) Obtain the relation for output power in induction motor. Also explain limitations in power calculation. (CO4, CO6)
4. (a) List the factors to be considered for the choice of specific electric loading in induction motor. (CO4, CO1)
- (b) Discuss the constructional details of an Induction motor with a suitable diagram. (CO4, CO1)
- (c) Obtain the following design formation for the stator of a 30 kW, 440 V 3Φ , 6 pole, 50 Hz delta connected, squirrel cage induction motor :
- (i) Main dimension of the stator, (ii) No. of turns/phase
(iii) No. of stator slots, (iv) No. of conductors per slot. Assume suitable values for the missing design data. (CO4, CO1)

(3)

5. (a) What is meant by specific electric and specific magnetic loading in a dc machine ? From first principles, obtain the expressions for specific magnetic loading and specific electric loading. (CO6, CO5)
- (b) Derive the output equation of DC machine. (CO6, CO5)
- (c) Determine the number of poles, armature diameter and core length for the preliminary design of a 500 kW, 400 V, 600 rpm, dc shunt generator assuming an average flux density in the air gap of 0.7 T and specific electric loading of 38400 ampere- conductors per metre. Assume core length/ pole arc = 1.1. Apply suitable checks. (CO6, CO5)

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UCE-501

**B. TECH. (ELECTRICAL ENGINEERING)
(FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

DISASTER MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define the term 'disaster' and describe its classification. (CO1)
(b) Explain the relationship of development and disaster. (CO1)
(c) Highlight key environmental concerns pertaining to disasters. (CO5)
2. (a) Describe structural and non-structural mitigation measures in disaster management. (CO2)
(b) Describe Hiroshima and Nagasaki event as a disaster. (CO6)
(c) Describe role of IT which could be used for disaster management. (CO5)

P. T. O.

(2)

3. (a) Highlight development perspective to disaster management with focus on disaster management in Himalayan regions. (CO4)
- (b) 'Sustainable management of natural resources is essential to provide livelihood and environmental security.' Discuss. (CO5)
- (c) Write a note on human activities as an event which leads to a huge disaster. (CO4)
4. (a) Write a detailed note on cyclone shelter as a preventive measure. (CO5)
- (b) Explain the relationship between hazard, risk, vulnerability and capacity. (CO2)
- (c) Discuss the role of NDMA, SDMA and DDMA in disaster management. (CO2)
5. (a) Write a note on Disaster Management Act, 2005. (CO1)
- (b) Write a case study on any one recent manmade disaster in world. (CO6)
- (c) Write the constraints of relief distribution. (CO6)